Keys verification

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1. Purpose & Scope

This procedure defines the method for determining the appropriate dimensions for keys and keyways used in torque transmission.

This procedure applies to horizontal and vertical pump shafts with torque transmitted through a key. Combined loads, such as axial thrust plus torque, are not considered.

This procedure is applicable to Nova Milanese plant and La Spezia plant.

2. Abbreviations

SCF	Stress Concentration Factor
b	key thickness [mm]
f	key fillet [mm]
h	key height [mm]
L_{key}	length of the key's straight section [mm]
D_h	hub diameter [mm]
D_j^*	reduced journal diameter [mm]
D_j	journal diameter [mm]
F_h	force generated by torque on key/hub's contact surface [N]
F_{sh}	force generated by torque key/shaft's contact surface [N]
H_h	height of key/hub contact surface [mm]
H_{sh}	height of key/shaft contact surface [mm]
K_a	application factor [-]
K_d	design factor [-]
K_f	fatigue life factor
K_{tA}	SCF for normal stress in point A [-]

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K_{tsA} SCF for shear stress in point A [-]

K_{tB} for normal stress in point B [-]

K_{tsB} SCF for shear stress in point B [-]

r fillet radius [mm]

R_{limit} hub (geometrical) limit radius [mm]

SF safety factor [-]

T nominal torque [Nm]

T_{design} design torque [Nm]

$T_{short\ circuit}$ short circuit torque [Nm]

σ_{yield} material yield stress [MPa]

$\sigma_{ultimate}$ material ultimate stress [MPa]

σ_{max} maximum normal stress [MPa]

τ_{max} maximum shear stress [MPa]

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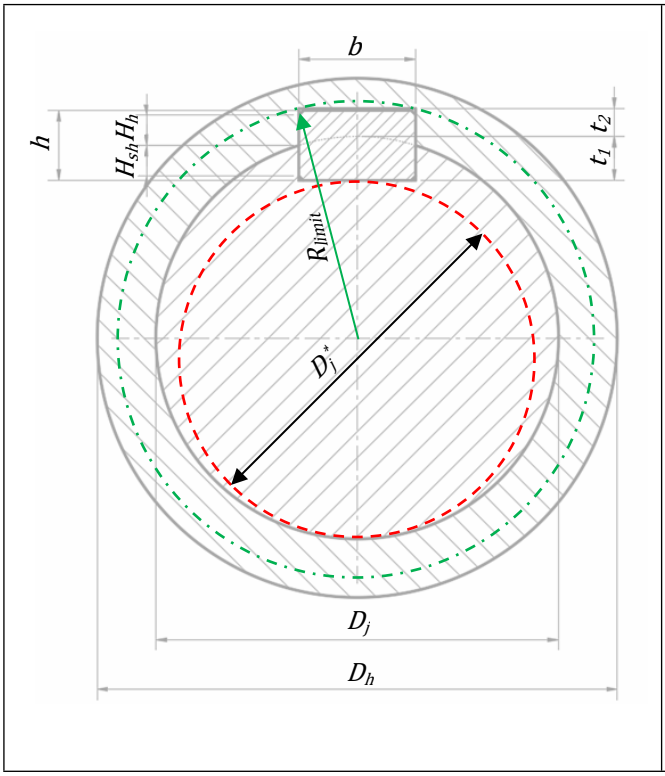
3. Key geometry

Figure 1 shows the geometry of the key and keyway, that affects torque distribution between transmission components. The standard dimensions of the keys are provided in "Annex A: key dimensions".

All criteria, excluding those related to stress concentration factors, are applicable for keys of type A, B, or A/B. The length of the key's straight section L_{key} should be considered for all calculations.

The hub diameter D_{hub} is subject to very stringent geometric limitations that cannot be overlooked: for the feasibility of machining, it must have a minimum margin of +20% over twice the value of R_{limit}.



$$H_{sh} = t_1 - \frac{D_h - \sqrt{D_h^2 - b^2}}{2} - f$$

$$H_h = h - 2f - H_{sh}$$

$$F_{sh} = \frac{2T_{design}}{\sqrt{D_h^2 - b^2} - H_{sh}}$$

$$F_h = \frac{2T_{design}}{\sqrt{D_h^2 - b^2} + H_h}$$

$$D_j^* = D_j - t_1$$

R_{limit} This formula is based exclusively on geometrical

$$. = \sqrt{\left(\frac{D_j}{2} + t_2\right)^2 + \frac{b^2}{4}}$$

Figure 1: Key geometry.

4. Material's mechanical properties

For the dimensioning procedure, both yield and ultimate stress are required for the shaft, key and hub materials.

5. Dimensioning procedure: nominal conditions

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5.1. Load factors

The dimensioning procedure is determined by torque transmission through the keys. *Table 1* summarizes multiplication factors, which vary according to the type of power source application. *Table 2* presents different connection designs, while *Table 3* provides the required fatigue life coefficients.

Table 1 Application factor K_a.

	Type of load			
Power source	Uniform (generator, fan)	Light Shock (oscillating pumps, etc.)	Intermittent Shock (actuating pumps, etc.)	Heavy Shock (punches, shears, etc.)
Uniform (turbine, motor)	1	1.2	1.5	1.8
Light shock (hydraulic motor)	1.2	1.3	1.8	2.1
Medium shock (internal combustion engine)	2	2.2	2.4	2.8

Table 2 Design factor K_d.

Fixed (close fit) loaded	No relative axial/radial movement of the hub and the shaft occurs; the relative positions of the shaft and hub are fixed using a suitable bearing or constructional design (e.g. press-fit, shaft nuts).	1
Flexible/Sliding - Unloaded	Relative positions of the shaft and hub are not fixed; axial movements of the hub along the shaft occur only when coupling is unloaded. The shaft is effectively fixed when loaded.	3
Flexible/Sliding loaded	Sliding coupling with loading: relative positions of the shaft and hub are not fixed; axial shifts of the hub along the shaft occur in loaded couplings.	9

Table 3 Fatigue life factor K_f.

Number of cycles [-]	Unidirectional	Fully reversed
1.00E+03	1.8	1.8
1.00E+04	1.0	1.0
1.00E+05	0.5	0.4
1.00E+06 (default value)	0.4	0.3
1.00E+07	0.3	0.2

Torque magnitude[1] considered for dimensioning at nominal conditions is equal to:

$$T_{design} = T \cdot \frac{K_a K_d}{K_f}$$

[1] When two keys are used, due to uneven load distribution, only 65% of the total torque should be considered for key dimensioning. However, this configuration is not ideal and should be avoided whenever possible.

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5.2. Stress verification components

Stress verification involves five verification steps:

1. Shaft minimum diameter.
2. Shear on key.
3. Pressure on shaft and key.
4. Pressure on hub and key.
5. Shaft stress with SCF (normal stress and shear stress at point A and B).

5.2.1. Shaft minimum diameter

The maximum shear stress on the reduced shaft must be less than or equal to the allowable shaft shear stress.

$$\tau_{max} = \frac{16 \cdot T_{design}}{\pi \left(D_j^*\right)^3} \leq \tau_{all} = \frac{\sigma_{shaf\ yield}}{\sqrt{3} \cdot SF}$$

5.2.2. Key shear stress

Key shear stress should be less than or equal to the allowable key shear stress:

$$\tau_{key} = \frac{F_{sh}}{b \cdot L_{key}} \leq \tau_{all} = \frac{\sigma_{key\ yield}}{\sqrt{3} \cdot SF}$$

5.2.3. Pressure on shaft and key

The contact pressure between the shaft and key must not exceed their allowable normal stress:

$$p_{sh} = \frac{F_{sh}}{\left(H_{sh} - f\right) \cdot L_{key}} \leq \sigma_{all} = \frac{min\left(\sigma_{key\ yield}; \sigma_{shaft\ yield}\right)}{SF}$$

5.2.4. Pressure on hub and key

The contact pressure between hub and key must not exceed their allowable normal stress:

$$p_h = \frac{F_h}{\left(H_h - f\right) \cdot L_{key}} \leq \sigma_{all} = \frac{min\left(\sigma_{key\ yield}; \sigma_{hub\ yield}\right)}{SF}$$

5.2.5. Shaft stress with SCF

This section of the verification procedure relies on the stress generated on the shaft where concentrators are present. Stress Concentration Factor (SCF) is determined for U.S.

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standard keyseats[2] (keyways). Note that the SCF can vary if these dimensions are not met, therefore this part is recommended but not mandatory if non-U.S. standard keyways are used.

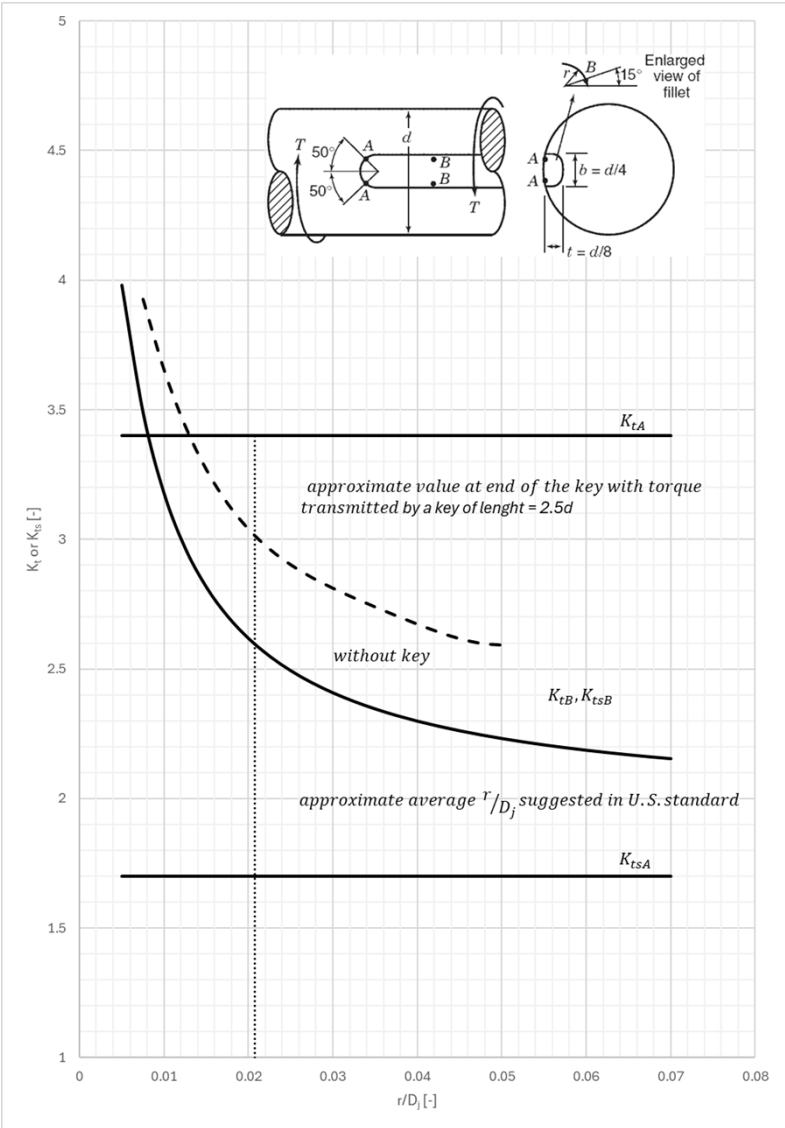

Figure 2 SCF.

The maximum normal stress, tangential to the semicircle, occurs at 50 degrees from the axial direction and is independent of the ratio r/d. The maximum shear stress is oriented at

2 $\frac{b}{D} = 1/4, \frac{h}{D} = 0.09, \frac{r}{D} = 1/48$

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45 degrees relative to the maximum normal stress. The corresponding shear Stress Concentration Factor (SCF) is half the value of the normal SCF:

$$K_{tA} = cost = 3.4$$

$$K_{tsA} = cost = 1.7$$

$$\sigma_{max\ A} = K_{tA} \cdot \frac{16 \cdot T_{design}}{\pi (D_j)^3} \leq \sigma_{all} = \frac{\sigma_{shaf\ yield}}{SF}$$

$$\tau_{max\ A} = K_{tA} \cdot \frac{16 \cdot T_{design}}{\pi (D_j)^3} \leq \tau_{all} = \frac{\sigma_{shaf\ yield}}{\sqrt{3} \cdot SF}$$

The maximum stresses in the fillet are present at point B in the straight part of the fillet. The maximum shear stresses are in the longitudinal and perpendicular directions. The maximum normal stresses are of the same magnitude and are at 45 degrees to the direction of the shear stress:

$$K_{tB} = K_{tsB}$$

$$\sigma_{max\ B} = \tau_{max\ B} = K_{tsB} \cdot \frac{16 \cdot T_{design}}{\pi (D_j)^3}$$

Normal and shear stress at point A and B of the shaft must be less or equal to the allowable shaft shear and normal stress:

$$\sigma_{max\ A} \leq \sigma_{all} = \frac{\sigma_{shaf\ yield}}{SF}$$

$$\tau_{max\ A} \leq \tau_{all} = \frac{\sigma_{shaf\ yield}}{\sqrt{3} \cdot SF}$$

$$\sigma_{max\ B} \leq \sigma_{all} = \frac{\sigma_{shaf\ yield}}{SF}$$

$$\tau_{max\ B} \leq \tau_{all} = \frac{\sigma_{shaf\ yield}}{\sqrt{3} \cdot SF}$$

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Mod. 1140 Keys Verification is available to perform the calculations.

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6. Dimensioning procedure: short circuit

6.1. Load factors

For short circuit conditions, the verification procedure follows a similar protocol. The torque amplitude is assumed to be ten times the nominal torque, and the allowable stress is calculated based on the ultimate stress rather than the yield stress.

The values presented in Table 4 are based on empirical evaluation. The maximum value occurs in the first milliseconds after the fault (impulsive torque). The torque can be asymmetric and oscillating, so the maximum effective value is often considered.

Table 4 Amplification factor for nominal torque.

	$T_{short\ circuit}/T$ [-]
Asynchronous motor	6 - 13
Synchronous motor	3 - 8
Generator	2 - 6

Short circuits should be considered anomalous conditions that seldom occur; therefore, the torque magnitude considered for dimensioning becomes:

$$T_{short\ circuit} = 10 \cdot K_d \cdot T$$

The applied verification formulae are identical to those outlined in the preceding sections, except for T_{design} substituted with $T_{short\ circuit}$, and the safety factor SF set to 1.

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7. References

Shingley's Mechanical Engineering Design – R. G. Budynas, J. K. Nisbett

Peterson's Stress Concentration Factor - W. D. Pilkey, D. F. Pilkey

When Splines Need Control - D. W. Dudley

The Stress Concentration in Keyways when Torque is Transmitted through Keys - H. Okubo, K. Hosono, K. Salaki

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8. Annex A: key dimensions

DN_{min} [mm]	DN_{max} [mm]	b [mm]	h [mm]	t_1 [mm]	t_2 [mm]	f [mm]	bxh
6	8	2	2	1.2	1	0.25	2x2
8	10	3	3	1.8	1.4	0.25	3x3
10	12	4	4	2.5	1.8	0.25	4x4
12	17	5	5	3	2.3	0.4	5x5
17	22	6	4	2.5	1.8	0.4	6x4
17	22	6	5	3	2.3	0.4	6x5
17	22	6	6	3.5	3.5	0.4	6x6
22	30	8	5	3	2.3	0.4	8x5
22	30	8	6	3.5	2.8	0.4	8x6
22	30	8	7	4	3.3	0.4	8x7
22	30	8	8	5	3.3	0.6	8x8
30	38	10	8	5	3.3	0.6	10x8
30	38	10	10	6	4.3	0.6	10x10
38	44	12	8	5	3.3	0.6	12x8
38	44	12	12	7.5	4.9	0.6	12x12
44	50	14	9	5.5	3.8	0.6	14x9
44	50	14	14	9	5.4	0.6	14x14
50	58	16	10	6	4.3	0.6	16x10
58	65	18	11	7	4.4	0.6	18x11
65	75	20	12	7.5	4.9	0.8	20x12
75	85	22	14	9	5.4	0.8	22x14
85	95	25	14	9	5.4	0.8	25x14
95	110	28	16	10	6.4	0.8	28x16

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DN_{min} [mm]	DN_{max} [mm]	b [mm]	h [mm]	t_1 [mm]	t_2 [mm]	f [mm]	bxh
110	*130*	*32*	*18*	*11*	*7.4*	*0.8*	*32x18*
130	*150*	*36*	*20*	*12*	*8.4*	*1.2*	*36x20*
150	*170*	*40*	*22*	*13*	*9.4*	*1.2*	*40x22*
170	*200*	*45*	*25*	*15*	*10.4*	*1.2*	*45x25*
200	*230*	*50*	*28*	*17*	*11.4*	*1.2*	*50x28*
230	*260*	*56*	*32*	*20*	*12.4*	*1.6*	*56x32*
260	*290*	*63*	*32*	*20*	*12.4*	*1.6*	*63x32*
290	*330*	*70*	*36*	*22*	*14.4*	*1.6*	*70x36*
330	*380*	*80*	*40*	*25*	*15.4*	*3*	*80x40*
380	*440*	*90*	*45*	*28*	*17.4*	*3*	*90x45*
440	*500*	*100*	*50*	*31*	*19.5*	*3*	*100x50*

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